

SMART INDIA HACKATHON 2026

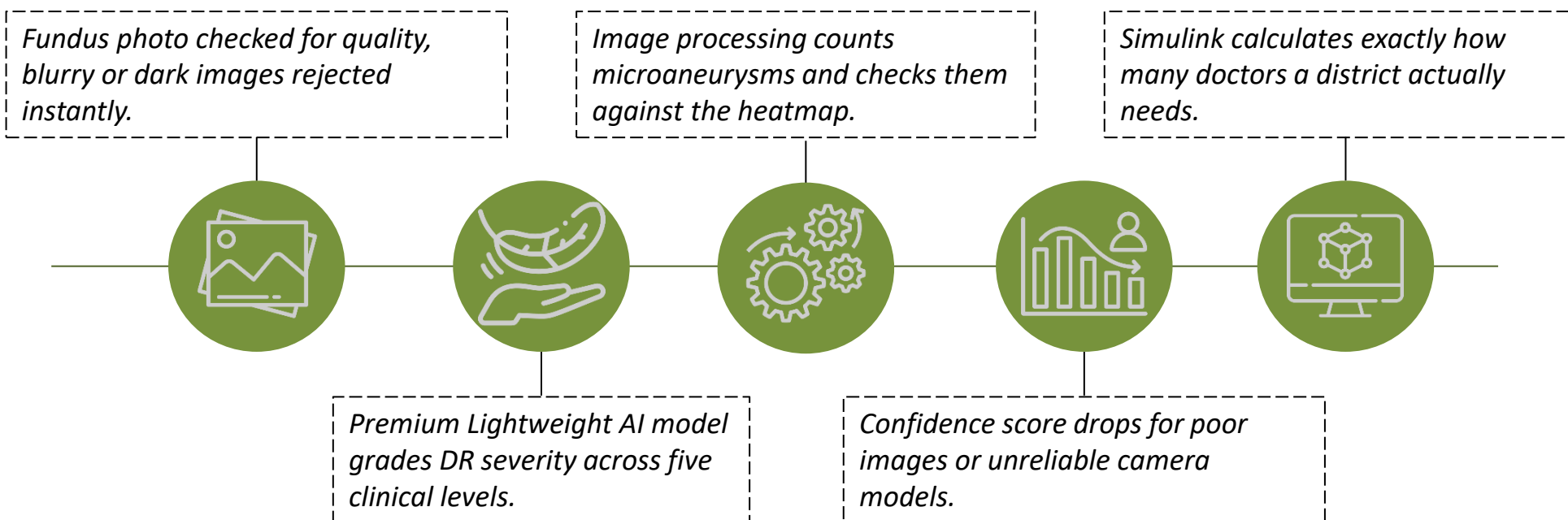


SMART INDIA
HACKATHON
2026

- **Problem Statement ID-** 26038
- **Problem Statement Title-** Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Theme-** MedTech/BioTech/HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name-** Ctrl Alt Elite



PROCESS OF THE SOLUTION



UNIQUENESS OF THE SOLUTION

CAMERA-AWARE TRUST

A lookup table flags specific camera models with known miss rates at the DR boundary, a check no device vendor makes about its own hardware.

LIVE STAFFING NUMBERS

Simulink outputs one number, minimum ophthalmologists needed for a 48-hour turnaround, not a queue chart left for someone else to interpret.



VENDOR-AGNOSTIC DESIGN

Wraps around whatever fundus camera a clinic already owns, no hardware lock-in the way Remidio and Forus require.

REFUSE-TO-GUESS SAFETY

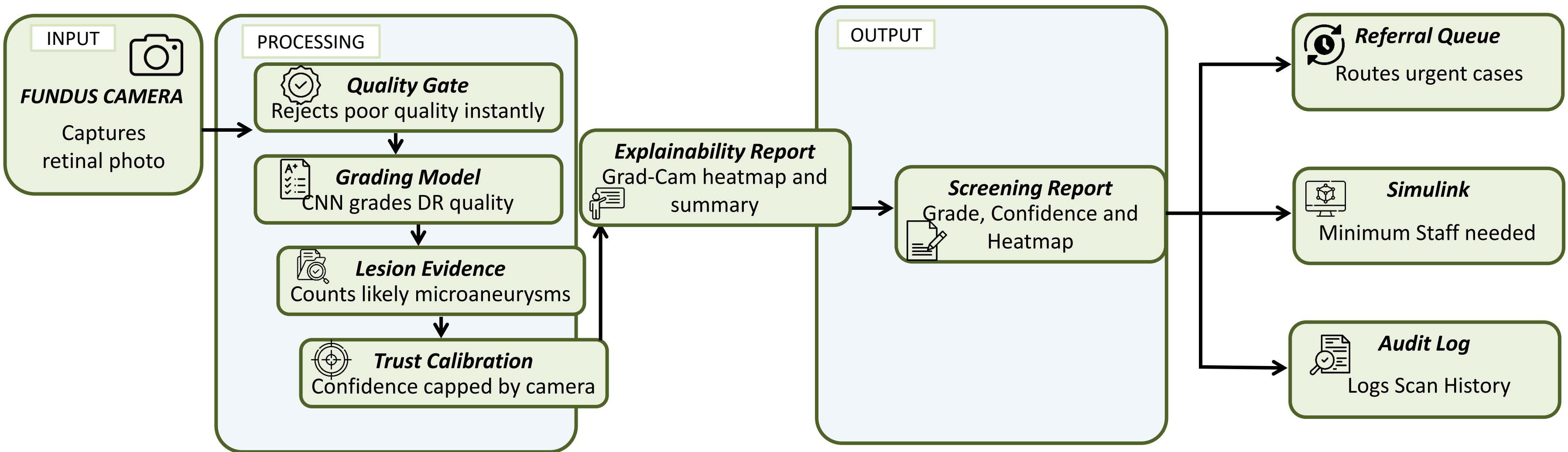
Grading stops when the model itself is uncertain, not only when the photo is unreadable, a second safety net competitors skip.

KEY FEATURES

- Works with any existing fundus camera, no hardware lock-in required.
- Flags cases for human review instead of forcing a risky guess.
- Names the exact camera model behind an unreliable reading.
- Runs on cheap, low-power hardware through a lightweight AI model.
- Turns staffing decisions into one live number, not a static chart.

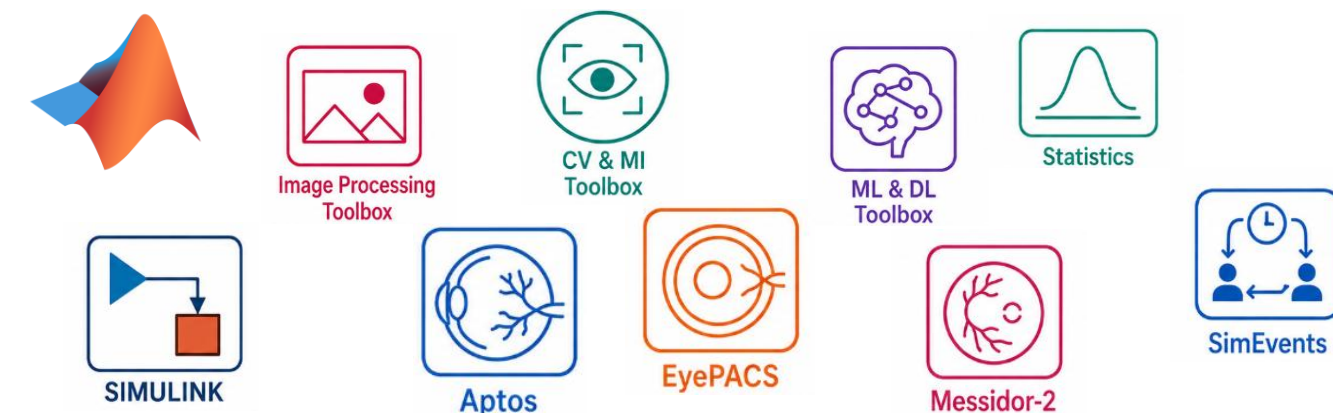
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M**Prevent
Vision Loss****Built Trust****Reach
Rural India****Support
Clinicians****Ensure
Precision**

TECHNICAL APPROACH



TECH STACK

- **Core Platform:** MATLAB, Simulink
- **Image Processing:** Image Processing toolbox, CV & MI Toolbox
- **AI and ML:** ML and DL Toolbox(MobileNet V3), Statistics
- **Datasets:** Aptos, EyePACS, IDRiD, Messidor-2
- **Deployment and Simulation:** SimEvents, Simulink



FEASIBILITY AND VIABILITY



FEASIBILITY

- **Preliminary Requirement:** Public fundus datasets already provide the labelled images needed for training and validation.
- **Market:** The AI-driven DR screening market is projected to reach USD 2.22 billion by 2033, growing at 21% annually.
- **Investment:** Built entirely on standard MATLAB and Simulink toolboxes, no custom hardware or GPU cluster required.



CHALLENGES

- **Image Quality:** Portable fundus cameras in field conditions produce blurry, poorly lit, or partial images that confuse naive classifiers.
- **Tiny Lesions:** Microaneurysms span only a handful of pixels and vanish under a standard image resize.
- **Clinical Trust:** Ophthalmologists need to validate AI output inside a workflow, not just accept a raw label.



APPLICATIONS

- **Primary Screening:** Grades DR severity from Level 0 to 4 at primary healthcare centres before referral.
- **Trust Calibration:** Flags unreliable readings tied to specific camera models before a doctor ever sees them.
- **Capacity Planning:** Tells a district exactly how many ophthalmologists it needs to clear a target patient volume.



GROWTH DRIVERS

- **AI Advancements:** MobileNetV3 matches heavier models at a fraction of the cost.
- **National Screening Push:** Diabetes burden and doctor shortage drive screening.
- **Government and NGO Support:** NHM, Aravind etc already fund rural camps.
- **Falling Camera Costs:** Cheaper fundus cameras widen the clinic pool needing this layer.



RISKS

- **Data Gaps:** Public datasets may not generalize well to Indian field images.
- **Vendor Pushback:** Camera makers may resist flagging their own hardware weaknesses.
- **Regulatory Scope:** CDSCO clearance may apply if scope moves past decision.
- **Incumbent Competition:** Remidio and Forus already have funding, clearance, and deployed hardware.



BUSINESS POTENTIAL

- **SaaS Licensing:** Per-district or per-camp-season subscriptions sold to state health departments and NGOs.
- **Vendor Integration Fees:** Wrap the trust-calibration layer around an existing camera vendor's AI output for a fee.
- **Outcomes Data:** Anonymized program-level analytics sold to health ministries and researchers after real deployments.

IMPACT AND BENEFITS

TARGET AUDIENCE

Health
DepartmentsOphthal-
mologistsCemera
VendorsPrimary
Health Centres

NGOS

ADVANTAGES

SOCIAL

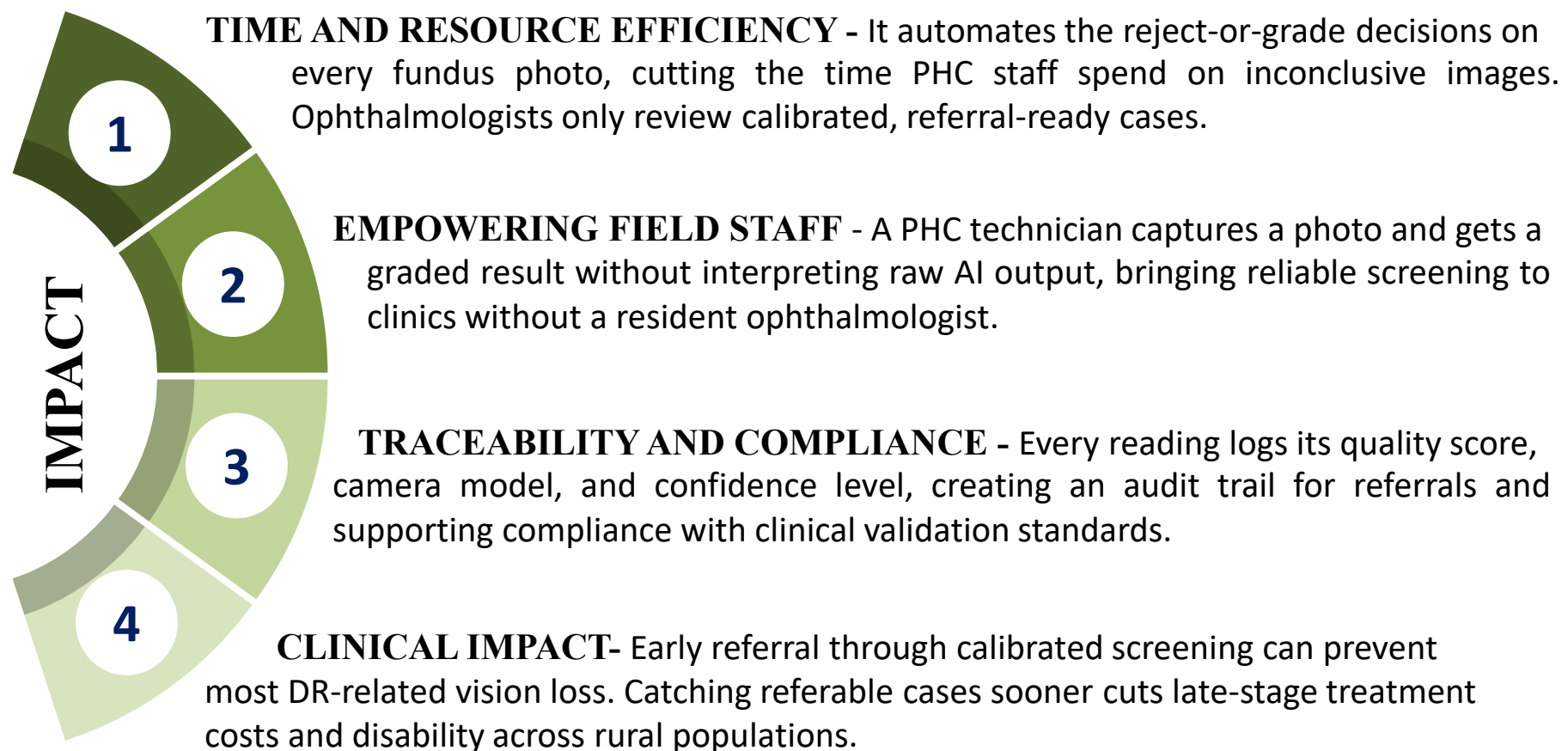
- **Improves access** by bringing screening to clinics without a resident ophthalmologist.
- **Builds patient trust** by explaining every reading instead of a black-box grade.
- **Reduces preventable blindness** by catching referable cases early across underserved populations.

CLINICAL

- **Improves diagnostic reliability** by tying confidence to image quality and camera history.
- **Reduces missed cases** through a refuse-to-guess safety net on uncertain readings.
- **Standardizes referral criteria** using validated severity grading and clinical evidence overlays.

ECONOMIC

- **Cuts staffing costs** by outputting the exact ophthalmologist count a district needs.
- **Lowers late-stage costs** by catching disease before advanced vision loss occurs.
- **Opens a revenue stream** through per-district SaaS licensing and vendor integration fees.



RESEARCH AND REFERENCES

REFERENCES

1.

[Sunderajah et al., "The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence," Nature Medicine, 2025.](#)

2.

[Sunderajah et al., "Developing specific reporting guidelines for diagnostic accuracy studies assessing AI interventions: The STARD-AI Steering Group," Nature Medicine, 2020.](#)

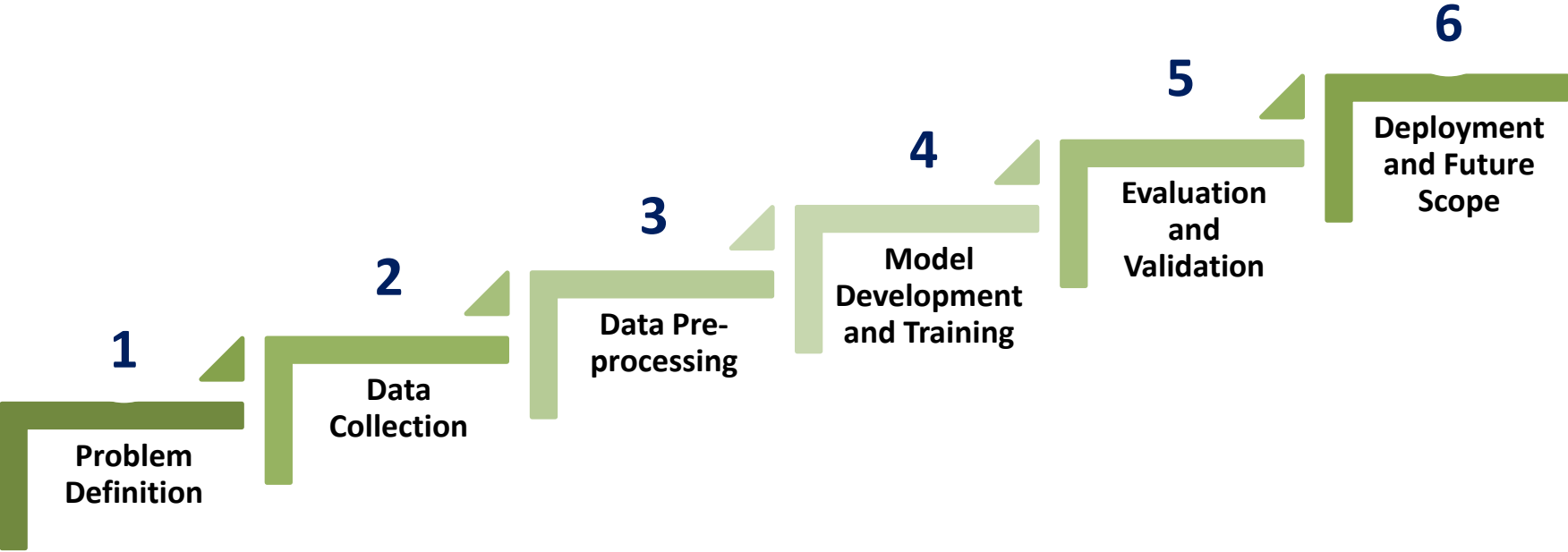
3.

["DECIDE-AI reporting guideline for early-stage clinical evaluation of AI-based decision-support systems," Nature Medicine, 2022.](#)

4.

[International Clinical Diabetic Retinopathy Disease Severity Scale, American Academy of Ophthalmology.](#)

RESEARCH WORKFLOW



HOW WE ARE DIFFERENT?

FEATURES	CURRENT MANUAL PROCESS	EXISTING AI DEVICES	CERTUS
Eliminates Manual Guesswork	✗	✓	✓
Reduces Errors	✗	≈	✓
Flags Unreliable Camera	✗	✗	✓
Refuses Low-Confidence Grades	✗	✗	✓
Works With Any Existing Camera	✗	≈	✓
Staffing and Capacity Planning	✗	✗	✓
Consistency and Accuracy	✗	≈	✓
Scalability and Expandability	✗	≈	✓